Yongwoo Son, DVM

Veterinarian · MS student, Wildlife Health Laboratory
College of Veterinary Medicine, Konkuk University, Seoul, Korea

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Interests Viral genomic epidemiology · Phylodynamics · Wildlife zoonoses · Arboviruses

Education

2025–present **MS, Veterinary Infectious Disease** (expected February 2027)
Wildlife Health Laboratory · Konkuk University, Seoul · GPA 4.5/4.5
Thesis: "Evolution and dispersal of avian paramyxoviruses in wild birds of the East Asian–Australasian Flyway"
Supervisor: Prof. Dong-hun Lee

2017–2025 **Doctor of Veterinary Medicine (DVM)**
College of Veterinary Medicine, Gyeongsang National University, Jinju · GPA 3.87/4.5

Ongoing Research

Viral genomic surveillance in Lao PDR and Tanzania
This project started with capacity-building workshops at the national public health laboratories of Lao PDR and Tanzania. For the workshops, I adapted the analysis pipeline to the primer schemes used in each country and to the computers available at each laboratory. I am now working with both laboratories on phylogeographic analysis of dengue virus sequences from the two countries, and of chikungunya and measles virus sequences from Lao PDR, to find out how these viruses move across borders and which lineages are circulating.

Intercontinental spread of avian paramyxoviruses
I investigated avian paramyxoviruses in AIV-negative wild bird samples using whole-genome sequencing. To understand how these viruses moved between continents and whether East Asia played a role in their spread, I now work on phylogeographic analysis of APMV-1 and APMV-4. Along the way, I also discovered a virulent Newcastle disease virus in a great cormorant and avian orthoavulavirus 16 in a cinereous vulture, a host in which it had not been reported before.

Avian influenza surveillance in wild birds
Participating in the national highly pathogenic avian influenza (HPAI) surveillance program, I collect fecal samples from wild birds across Korea and sequence whole genomes of the viruses detected.

Publications

2026 **Son Y**, Son Y, Kim D, Kim JY, Lee SH, Cho H, Jeong H, Song CS, Lee DH. Isolation and Genetic Characterization of a Virulent Newcastle Disease Virus Subgenotype XIII.1.2 from a Great Cormorant (*Phalacrocorax carbo*) in South Korea, 2026. Under review.

Araujo AC, Oliveira LM, **Son Y**, Kim JY, de Lima RR, Scagion GP, Caleiro GS, de Araújo J, Rosa RSM, Arns CW, Cordeiro IM, Dias PRBL, Spilki FR, Lee DH, Ferreira HL. Emergence of the GI-23 lineage of infectious bronchitis virus in Brazilian backyard poultry. *Avian Pathology*. doi:10.1080/03079457.2026.2718378

Kim JY, Lee DY, Otgontogtokh N, Myagmarsuren T, Seo YR, **Son Y**, Lee H, Park S, Lee SH, Tseren-Ochir EO, Lee DH, Song CS. Surveillance and Whole-Genome Analysis of Low Pathogenic Avian Influenza Viruses from Wild Birds in Mongolia, 2025. *Avian Diseases*, 70(3), 343–348. doi:10.1637/aviandiseases-d-26-00023

Son Y*, Lee SH*, Lee DY, Cho H, Son Y, Kim D, Song CS, Lee DH. Isolation and whole genome sequencing of an Avian orthoavulavirus 16 from cinereous vulture (*Aegypius monachus*) in South Korea, 2023. *Frontiers in Veterinary Science*, 13, 1739779. doi:10.3389/fvets.2026.1739779 (*Equal contribution)

Kim J*, **Son Y***, Hwang B, Kim Y, Kim J, Min W, Kim WH. Black-tailed Gull (*Larus crassirostris*): A Novel Host for *Trichomonas gypaetinii* with High Prevalence of Avian Trichomonads in South Korea. *Journal of Wildlife Diseases*, 62(1), 192–197. doi:10.7589/JWD-D-25-00039 (*Equal contribution)

2025 Ahn SM, Son S, **Son Y**, Lim SJ, Kim D, Choi KS, Kwon HJ. An *rpoB* Sequence Type Network as a Framework for the Evolutionary Investigation of *Clostridium perfringens*. *Microorganisms*, 13(12), 2768. doi:10.3390/microorganisms13122768

El Zowalaty ME, Taylor LJ, **Son Y**, Lee H, Rubrum AM, Webby RJ, Bustin SA, Young SG, Lee S-H, Lee D-H, Frieman MB. Discovery, phylogenetic, and comparative genomic analysis of novel avian gammacoronaviruses identified in feral pigeons (*Columba livia domestica*). *Journal of Virology*, 99(9), e01112-25. doi:10.1128/jvi.01112-25

2024 **Son Y***, Jin YB*, Cho EJ, Park AR, Flores RA, Nguyen TB, Lee SY, Altanzul B, Park KI, Min W, Kim WH. Comparative Analysis of Antibiotic Resistance and Biofilm Characteristics of Two Major *Enterococcus* Species from Poultry Slaughterhouses in South Korea. *Veterinary Sciences*, 11(4), 180. doi:10.3390/vetsci11040180 (*Equal contribution)

Flores RA, Cammayo-Fletcher PLT, Nguyen BT, Villavicencio AGM, Lee SY, **Son Y**, Kim JH, Park KI, Yoo WG, Jin YB, Min W, Kim WH. Genetic Characterization and Phylogeographic Analysis of the First H13N6 Avian Influenza Virus Isolated from Vega Gull in South Korea. *Viruses*, 16(2), 285. doi:10.3390/v16020285

2023 Kim CH*, **Son Y***, Choi YJ, Ko BH, Kang WH, Kim GA, Lee S, Kim WH. The prevalence of viral diseases in wild boars (*Sus scrofa*) in Gyeongsangnam-do, South Korea. *Korean Journal of Veterinary Service*, 46(1), 59–66. doi:10.7853/kjvs.2023.46.1.59 (*Equal contribution)

2022 **Son Y***, Cho S*, Ji JM, Cho JK, Bang SY, Choi YJ, Kim CH, Kim WH. Prevalence study of bovine viral diarrhea virus (BVDV) from cattle farms in Gyeongsangnam-do, South Korea in 2021. *Korean Journal of Veterinary Service*, 45(3), 211–219. doi:10.7853/kjvs.2022.45.3.211 (*Equal contribution)

Presentations

2026 **Listing status, host distribution, and human notifications for 30 zoonotic viruses in Korea, 2020–2025**
Korean Society for Zoonoses Autumn Conference, Busan · September 21 · Poster

Genomic Surveillance of Avian Paramyxoviruses Recovered from Avian Influenza-Negative Wild Bird Samples in Korea
2nd Joint Symposium of the Korea-Japan Network for the Control of Avian Influenza, Sapporo · August 28 · Oral

Whole-genome analysis of avian paramyxovirus 4 reveals two distinct Eurasian lineages
Croucher Summer Course: Virus Evolution, The University of Hong Kong · June 29–July 3 · Poster

2025 **Occupational Zoonotic Transmission of Virulent Newcastle Disease Virus in Nigeria, 2023**
Korean Society for Zoonoses Autumn Conference, Seoul · November 21 · Poster

2023 **Assessment of Viral Diseases in Wild Boars of Gyeongsangnamdo, South Korea: A Surveillance Study**
16th Asian Society of Conservation Medicine (ASCM) Conference, Jeju · October · Poster

Grants and Scholarships

2025–2026 **Master's Student Research Grant**
National Research Foundation of Korea · KRW 12,000,000 · Principal Investigator · RS-2025-25421559
“Disease Ecological Analysis of the Evolution and Transmission Dynamics of Avian Paramyxoviruses in Wild Birds”

2024 **Research Scholarship**
The Health Fellowship Foundation · KRW 5,000,000
“Epidemiological Analysis of Zoonotic Pathogens from Wild Birds in Korean National Parks”

Research Experience

2026 **Trainer, Genomic Epidemiology Fellowship Program, Tanzania**
KOFIH–NPHL, Dar es Salaam
Ran a training with NPHL scientists on sequencing archived dengue samples and a phylogeographic analysis of Marburg virus isolates.

Trainer, Viral Genome Sequencing & Bioinformatics, Lao PDR
KCDC–SHEPHERD–NCLE, Vientiane
As part of a sentinel site surveillance program for acute febrile illness, sequenced dengue, chikungunya and measles virus genomes using tiling-amplicon primers.

2024 **Research Intern**
Avian Disease Laboratory, Seoul National University
Learned avian influenza virus cloning and *rpoB*-based sequence typing of *Clostridium perfringens*.

2020–2025 **Undergraduate Researcher**
Laboratory of Veterinary Infectious Diseases, Gyeongsang National University
Worked on avian coccidiosis and ran surveillance of viruses in various animals.

Veterinary Experience

2023 **Southern Conservation Center, Korea National Park Service**
Assisted with artificial insemination trials for the Asiatic black bear restoration program and with treating wild animals.

Cetacean Research Institute, National Institute of Fisheries Science
Necropsied dolphins and seals and screened for parasites and avian influenza virus.

2022 **Comparative Pathology Laboratory, Asan Medical Center**
Learned basic mouse histopathology and examined lung sections from drug toxicity studies.

2022 **National Institute of Wildlife Disease Control and Prevention**
Attended a training on the national diagnostic and response protocols for avian influenza and African swine fever.

2021 **WOAH Reference Laboratory for Brucellosis, Animal and Plant Quarantine Agency**
Trained in *Brucella* diagnostics, MLVA and MLST typing.

2020 **Jeonnam Wildlife Rescue Center**
Helped treat and rehabilitate rescued wildlife.

Skills

Laboratory Molecular diagnostics, cell culture, NGS (Illumina), in vivo experiments in chickens and disease diagnostics in poultry

Analysis Phylogenetic and phylodynamic analysis based on BEAST X